Decodability Is Not Localization: Auditing Hidden-State Error Probes in Qwen2.5-Math-1.5B-Instruct

Anonymous Author(s)
Affiliation
Address
email

Abstract

We audit what a hidden-state error probe supports in a mathematical reasoning trace. In a case study of Qwen2.5-Math-1.5B-Instruct on ProcessBench, a linear probe predicts the persistent target *invalid so far* with held-out AUROC 0.866 [95% CI $[0.849, 0.884]$]. Equally tuned controls reach AUROC 0.751 with prefix TF-IDF, 0.776 with structural metadata, and 0.810 with both. A diagnostic model that also receives final-answer correctness reaches 0.874, but that field is a benchmark annotation unavailable at inference time. A post-hoc nested comparison shows that adding the full selected-layer hidden representation to the text and metadata model improves AUROC by 0.0587 [95% CI $[0.0414, 0.0745]$], log loss by -0.0798 [95% CI $[-0.1038, -0.0548]$], and Process F1 by $+0.1866$ [95% CI $[0.1397, 0.2340]$] on the inspected split. The probe does not reliably localize the first error: exact localization is 28.9%, correct-trace rejection is 61.2%, and detected errors occur 0.59 steps late on average. A post-hoc matched transition probe reaches AUROC 0.769 on 380 held-out pairs, but placebo reuse and covariate imbalance make this result exploratory. The revised behavioral readout has AUROC 0.342, so it fails the intervention validity gate. The evidence supports incremental predictive information in this case study. It does not establish generalization, calibration, self-monitoring, localization, or causal use.

1 Introduction

Process supervision evaluates the validity of intermediate steps rather than only the final answer (Cobbe et al., 2021; Lightman et al., 2023). This creates a representational question: does a model's hidden state change when a written solution has become invalid? A probe can answer a narrow predictive version of that question. It cannot, by itself, show that the model uses the decoded feature, nor that a threshold crossing identifies the first error. With a persistent label, every boundary after an error is positive. A classifier can then use accumulated inconsistency, position, solution length, generator identity, or final-answer outcome without responding at the onset.

We study these distinctions in one bounded setting. The model is Qwen2.5-Math-1.5B-Instruct and the data are 3,400 model-generated traces from ProcessBench (Yang et al., 2024; Zheng et al., 2025). The analysis uses problem-grouped partitions, a persistent invalid-so-far target, hidden-state probes, visible text and metadata controls, temporal randomization, matched transitions, a natural-token boundary control, and a gated behavioral intervention. The analysis was designed as an audit of evidence levels rather than as a claim that one score is a complete verifier.

The results have a specific shape. Hidden states predict the persistent target. They also add information to a nuisance model on the inspected split when compared with the same model augmented by the selected hidden representation. The score has temporal correspondence with the annotated onset, but its threshold gives modest exact localization and many false alarms on correct traces. The

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

behavioral readout fails its AUROC gate, so the intervention is not interpreted as evidence of causal use.

2 Related work and contribution boundary

Process reward models learn step quality from supervision on intermediate reasoning (Lightman et al., 2023; Li et al., 2023). ProcessBench provides annotations of the earliest error in mathematical reasoning traces and draws from GSM8K, MATH, OlympiadBench, and Omni-MATH (Zheng et al., 2025; Hendrycks et al., 2021; He et al., 2024; Gao et al., 2024). We use these sources for one representation analysis. They are not independent datasets, and the analysis does not test model-family generalization.

Several recent studies use internal states to score reasoning. Sun et al. probe arithmetic errors and use the result for selective re-prompting (Sun et al., 2025). ReProbe studies hidden-state step scoring for test-time scaling, and STEP uses a hidden-state scorer to prune reasoning traces (Ni et al., 2026; Liang et al., 2026). Alvarez and Baheri study first-error detection with hidden-state transport geometry, including cross-model and cross-dataset evaluations (Alvarez and Baheri, 2026). Yuan et al. separate diagnostic error awareness from causal use in a multi-model study (Yuan et al., 2026). Our contribution is narrower. We report an audit of one model and one benchmark construction that separates pooled prediction, first crossing, temporal onset correspondence, source transfer, conditional prediction, and a behavioral validity gate.

Probing literature distinguishes decodability from the information used by a model (Hewitt and Liang, 2019; Belinkov, 2022; Elazar et al., 2021). Our intervention therefore has a precondition: the unmodified behavioral readout must distinguish the relevant baseline states. The paper reports the intervention records and the failed precondition without interpreting a dose response as causal evidence.

3 Experimental design

3.1 Data, model, and partition

Each ProcessBench trace contains a problem, generator identity, source, written steps, final-answer correctness, and a human first-error index e_i. We set $e_i = -1$ for a fully correct trace. Traces with the same normalized problem text are assigned to the same partition. Seed 42 creates train, validation, and test partitions with target fractions of 60%, 20%, and 20%. The completed data flow retains 3,360 of 3,400 attempted traces and 24,909 of 25,697 attempted step boundaries. The 40 excluded traces exceed the 2,048-token context limit. The retained data contain 2,187 erroneous traces and 1,173 fully correct traces. The held-out partition contains 669 traces, 432 erroneous traces, 237 fully correct traces, and 4,985 step boundaries.

The retained test sources contain 84 GSM8K traces, 204 MATH traces, 188 OlympiadBench traces, and 193 Omni-MATH traces. The model record fixes Qwen2.5-Math-1.5B-Instruct, bfloat16 computation, a 2,048-token limit, and the dataset fingerprint `447f0a4b35c5747a9f9a3dab1e70d43f71efd501497b8cec668b71337099784a`. The completed extraction used one NVIDIA A100 GPU with batch size 16. The model reads each fixed trace in one causal forward pass and does not produce the traces.

3.2 Target and hidden-state probe

At boundary k of trace i, the primary target is

$$y_{ik} = \mathbb{1}[e_i \geq 0 \ \wedge \ k \geq e_i]. \tag{1}$$

This target means invalid so far. It labels the first invalid step and all later boundaries in an erroneous trace. It does not identify whether the current step is locally incorrect.

The extraction records a 1,536-dimensional residual-stream state at each written-step boundary and at 29 hidden-state indices. At each index, a standardized L2 logistic probe is trained with class balancing. The regularization grid is $C \in \{0.01, 0.1, 1, 10\}$. Validation AUROC selects C and validation Process F1 selects a threshold on a grid from 0.05 to 0.95 in increments of 0.005. Validation selects index 23, $C = 0.01$, and threshold 0.645. The selected probe is refit on train and validation data before the single test evaluation.

The predicted first error is the first threshold crossing, or -1 when no crossing occurs. We report boundary AUROC, average precision, step F1, error exactness among erroneous traces, correct-trace rejection, Process F1, complete-trace accuracy, detection rate, and error distance. Primary probe intervals use 1,000 whole-trace bootstrap draws. The later CPU control audit uses 2,000 draws. The conditional hidden-state comparison uses 1,000 draws. These analyses are reported as post-hoc audits rather than as one preregistered protocol.

3.3 Visible controls and conditional comparisons

The control analysis gives each nuisance model the same selection budget used for the probe. It selects C by validation AUROC, selects a threshold on validation data, refits on train and validation data, and uses inverse boundary-count weights during fitting so each trace contributes equal total training loss. Prefix TF-IDF uses the problem and reasoning text through the current boundary. Structural metadata uses source, generator, absolute and fractional position, trace length, and token count. The joint nuisance model combines prefix text and structural metadata.

Final-answer correctness is a benchmark annotation. It is available in the stored trace metadata, but it is not an online signal for a detector operating before the final outcome is known. We use models containing this field only as diagnostic upper bounds on nuisance predictability.

To test incremental prediction, we fit five feature sets. N contains prefix TF-IDF and structural metadata. $N+H$ adds the full selected-layer hidden representation. O contains the nuisance features and final-answer correctness. $O + H$ adds the full selected-layer hidden representation to O. H contains the hidden representation alone. Every feature set uses the same train, validation, and test partitions, the same regularization grid, trace-equal training weights, and refit on train plus validation. Paired intervals resample whole traces 1,000 times.

3.4 Post-hoc temporal and onset analyses

The temporal analyses were designed after inspecting the primary test result. We circularly shift each frozen score trajectory 5,000 times, preserving its values while breaking its alignment to the human annotation. We also compare the score jump at a noninitial first error with a transition in a fully correct trace matched by source, generator, relative position, trace length, and token count.

The transition probe uses the hidden-state difference $h_{i,e_i} - h_{i,e_i-1}$. The fixed partitions contain 1,152 train pairs, 387 validation pairs, and 380 test pairs. Across all partitions there are 1,919 pairs and 687 unique placebo traces. Placebo reuse has median 2, 90th percentile 6, and maximum 46 pairs per trace. The standardized differences are 0.011 for relative position, 0.146 for trace length, and 0.285 for token count. Layer and C are selected by validation AUROC, giving index 21 and $C = 0.01$. The test partition was inspected when this analysis was conceived, so the result is exploratory.

A boundary-location control compares the last natural token of each written step with the last token of its artificial step marker. It fixes index 23 and $C = 0.01$, fits thresholds on validation data, and evaluates on the original test partition. A paired whole-trace bootstrap compares the natural-token and marker-token results.

3.5 Behavioral validity gate

The intervention moves the boundary state along the learned probe direction at layer 23 over $\alpha \in \{-4, -2, -1, 0, 1, 2, 4\}$, where the scale is the training projection standard deviation. It also evaluates 20 random directions orthogonal to the learned direction at the two extreme values. The behavioral score is the difference between the conditional log probabilities of the answers "Yes" and "No" for a question asking whether the reasoning contains an error.

The pipeline interprets a learned-direction effect only when the unmodified readout has AUROC above 0.5 and nonzero specificity. The revised readout has specificity 0.203 but AUROC 0.342, so it fails this gate. The intervention records are therefore reported as an assay diagnostic, not as a causal test of model behavior.

4 Results

4.1 Hidden states add information to a nuisance model

The selected hidden probe reaches test AUROC 0.866 [95% CI $[0.849, 0.884]$], average precision 0.843 [95% CI $[0.813, 0.870]$], and step F1 0.743 [95% CI $[0.716, 0.769]$]. It reaches error exactness 0.289, correct-trace rejection 0.612, Process F1 0.393, and complete-trace accuracy 0.404.

The equally tuned nuisance models reach AUROC 0.751 for prefix TF-IDF, 0.776 for structural metadata, and 0.810 for their joint model. The outcome-augmented metadata model reaches 0.854, and the outcome-augmented joint model reaches 0.874. The latter uses final-answer correctness, so it is not a deployable baseline.

Table 1: Held-out predictive comparison. Nuisance controls receive the same validation selection budget and trace-equal training weights. The final-answer field is a diagnostic annotation and is not available to an online detector.

Predictor	AUROC	Proc. F1	Error exact	Corr. rej.
Hidden state	.866	.393	.289	.612
Prefix TF-IDF	.751	.274	.192	.477
Structural metadata	.776	.168	.148	.194
Joint text + metadata	.810	.210	.169	.278
Metadata + final outcome	.854	.262	.164	.646
Joint + final outcome	.874	.294	.176	.895

Table 2: Nested nuisance and hidden comparison. Δ is the hidden-augmented model minus its nuisance-only counterpart. All models use the same selection budget and refit protocol.

Nuisance set	Model	AUROC	Log loss	Proc. F1	Δ AUROC
N	N only	.811	.533	.211	N/A
	$N + H$	.869	.454	.397	+.0587
O	O only	.874	.440	.284	N/A
	$O + H$	.909	.385	.444	+.0348

The paired 95% AUROC intervals are $[0.0414, 0.0745]$ for $N + H - N$ and $[0.0242, 0.0451]$ for $O + H - O$. The O model includes final-answer correctness.

The conditional comparison gives the direct test of incremental prediction. N alone reaches AUROC 0.811, log loss 0.533, and Process F1 0.211. $N + H$ reaches AUROC 0.869, log loss 0.454, and Process F1 0.397. The paired changes are AUROC $+0.0587$ [95% CI $[0.0414, 0.0745]$], log loss -0.0798 [95% CI $[-0.1038, -0.0548]$], and Process F1 $+0.1866$ [95% CI $[0.1397, 0.2340]$]. The oracle comparison gives O AUROC 0.874, log loss 0.440, and Process F1 0.284. $O + H$ reaches AUROC 0.909, log loss 0.385, and Process F1 0.444. Its AUROC change is $+0.0348$ [95% CI $[0.0242, 0.0451]$]. The hidden AUROC advantage over the outcome-augmented joint nuisance model is -0.008 [95% CI $[-0.028, 0.013]$], while the hidden probe has a Process F1 advantage of 0.099 [95% CI $[0.040, 0.158]$] and an error-exactness advantage of 0.113 [95% CI $[0.064, 0.163]$]. The hidden probe has lower correct-trace rejection by 0.283 [95% CI $[-0.347, -0.218]$]. Equal trace weighting at evaluation changes AUROC by $+0.0013$ [95% CI $[-0.0069, 0.0103]$]. The corresponding changes are -0.0052 for average precision and -0.0087 for step F1, with intervals that include zero. The length-aware threshold analysis reduces Process F1 from 0.393 to 0.377 and correct-trace rejection from 0.612 to 0.540; error exactness is unchanged at 0.289.

4.2 Localization is weaker than pooled discrimination

The hidden score localizes 28.9% of erroneous traces exactly, 59.3% within one step, and 72.7% within two steps. It crosses on 88.7% of erroneous traces and falsely crosses on 38.8% of fully correct traces. Among detected erroneous traces, the mean signed localization error is $+0.59$ steps [95% CI $[+0.36, +0.80]$]. The score therefore tends to identify a persistent invalid state later than the annotated onset. The temporal randomization test gives exact localization 0.289 for the observed trajectories and a shifted-null mean of 0.164 [95% null interval $[0.137, 0.192]$], with permutation $p = 0.0002$. The observed within-one-step rate is 0.593, compared with shifted-null mean 0.436 and $p = 0.0002$. The observed Process F1 is 0.393, compared with shifted-null mean 0.259 and $p = 0.0002$. These shifts break annotation alignment but do not remove position as an explanation. The error-aligned score rises from 0.341 immediately before the annotated error to 0.559 at the onset, 0.680 one step later, and 0.731 two steps later. The matched correct-trace transition rises by 0.113, while the error-onset transition rises by 0.257. Their paired difference is 0.144 [95% CI $[0.096, 0.193]$]. Correct traces also show positive score drift. The result is consistent with a gradual change near the annotation, not a sharp threshold event. The natural-token boundary control gives AUROC 0.868 and Process F1 0.419. The marker-token control gives AUROC 0.866 and Process F1 0.404. Natural-token minus marker-token differences are $+0.002$ for AUROC [95% CI $[-0.007, +0.011]$] and $+0.026$ for Process F1 [95% CI $[-0.020, +0.075]$]. Both intervals include zero.

4.3 Matched onset transitions are exploratory

The transition probe reaches AUROC 0.769 [95% CI $[0.713, 0.820]$], average precision 0.741 [95% CI $[0.672, 0.816]$], and paired accuracy 0.753 [95% CI $[0.676, 0.824]$] on 380 test pairs. The position, destination-step TF-IDF, and shuffled-label controls reach AUROC 0.631, 0.671, and 0.585. The transition difference is therefore linearly distinguishable from the selected controls on the inspected split.

The interpretation is limited by the matching diagnostics. One placebo trace supports as many as 46 pairs. The matched groups are close on relative position but differ more on trace length and token count, with standardized differences 0.146 and 0.285. One-to-one matching and inverse-reuse sensitivity refits are implemented but no completed sensitivity artifact is present. We therefore treat this result as exploratory and do not use it to claim a general first-error mechanism.

4.4 Ranking transfers across ProcessBench sources

Source-specific probes trained on one source and evaluated on another have mean off-diagonal AUROC 0.805 and mean diagonal AUROC 0.845. Every off-diagonal AUROC is above 0.739. Thresholded Process F1 is less stable, with mean off-diagonal value 0.261, mean diagonal value 0.336, and off-diagonal values from 0.055 to 0.371. These are cross-source results within one benchmark construction. They do not test another dataset, model family, or model size.

4.5 Behavioral readout and intervention status

The original single-token CORRECT versus INCORRECT readout has AUROC 0.283 and specificity zero. The revised conditional Yes versus No readout has AUROC 0.342 and specificity 0.203 on 256 balanced boundaries. It passes the specificity condition but fails the AUROC condition required by the intervention pipeline. The learned-direction dose records therefore do not provide evidence for or against causal use of the decoded feature. The intervention section is empty by design after an invalid baseline readout, not because a valid behavioral test found a null effect.

5 Scope and limitations

The strongest supported statement is conditional: on one frozen split of one ProcessBench construction, hidden states add held-out predictive information to a tuned nuisance model. This statement is about prediction of the stored invalid-so-far label. It is not a claim that the model has a deployable error detector, that the hidden representation is unavailable from visible text, or that the model uses the feature in its own verdict.

The study uses one model, one benchmark construction, and one fixed partition. The post-hoc audits were designed after the primary result. Their intervals describe resampling uncertainty within the completed pipeline, not sensitivity to a new split, model, or selection procedure. The persistent target labels all later boundaries after an annotated error. The visible controls do not cover every possible textual or generator shortcut. The final-answer field is unavailable online. The source transfer analysis measures ranking, while threshold-dependent metrics vary more strongly. Probability calibration was not tested as a separate research question.

The transition analysis excludes 52 test traces whose first error is at step 0 and uses reused placebo traces. The natural-token control and temporal randomization were also post-hoc. Counterfactual activation patching has no result because drafted corrections were not independently verified and its behavioral readout did not pass the validity gate.

The detector is not suitable for grading, certification, or unattended feedback. On fully correct traces, the selected threshold produces a 38.8% false-alarm rate. Exact localization among erroneous traces is 28.9%, and the mean detected error is late. The stored results support use as a research diagnostic paired with human or formal verification only.

6 Conclusion

In Qwen2.5-Math-1.5B-Instruct reading ProcessBench traces, invalid-so-far prefixes are linearly decodable. The nested comparison shows an incremental held-out gain from adding the selected hidden representation to prefix text and structural metadata on the inspected split. The gain is not a complete verifier: localization remains weak, correct-trace rejection is limited, threshold behavior varies across sources and lengths, and the native behavioral readout fails its AUROC gate. The appropriate claim is therefore incremental predictive information in a bounded case study, not localization, self-monitoring, generalization, calibration, or causal use.

References

Tyler Alvarez and Ali Baheri. Where does reasoning break? step-level hallucination detection via hidden-state transport geometry. *arXiv preprint arXiv:2605.13772*, 2026. URL https://arxiv.org/abs/2605.13772.

Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022. doi: 10.1162/coli_a_00422.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. URL https://arxiv.org/abs/2110.14168.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021. doi: 10.1162/tacl_a_00359. URL https://aclanthology.org/2021.tacl-1.10/.

Bofei Gao, Feifan Song, Zhe Yang, Zefan Cai, Yibo Miao, Qingxiu Dong, Lei Li, Chenghao Ma, Liang Chen, Runxin Xu, Zhengyang Tang, Benyou Wang, Daoguang Zan, Shanghaoran Quan, Ge Zhang, Lei Sha, Yichang Zhang, Xuancheng Ren, Tianyu Liu, and Baobao Chang. Omni-MATH: A universal olympiad-level mathematics benchmark for large language models. *arXiv preprint arXiv:2410.07985*, 2024. URL https://arxiv.org/abs/2410.07985.

Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu, Xu Han, Yujie Huang, Yuxiang Zhang, Jie Liu, Lei Qi, Zhiyuan Liu, and Maosong Sun. Olympiad-Bench: A challenging benchmark for promoting AGI with olympiad-level bilingual multimodal scientific problems. *arXiv preprint arXiv:2402.14008*, 2024. URL https://arxiv.org/abs/2402.14008.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. *arXiv preprint arXiv:2103.03874*, 2021. URL https://arxiv.org/abs/2103.03874.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 2733–2743. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1275. URL https://aclanthology.org/D19-1275/.

Yifei Li, Zeqi Lin, Shizhuo Zhang, Qiang Fu, Bei Chen, Jian-Guang Lou, and Weizhu Chen. Making language models better reasoners with step-aware verifier. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 5315–5333. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.acl-long.291. URL https://aclanthology.org/2023.acl-long.291/.

Zhixiang Liang, Beichen Huang, Zheng Wang, and Minjia Zhang. Hidden states as early signals: Step-level trace evaluation and pruning for efficient test-time scaling. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 26800–26813. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.findings-acl.1336. URL https://aclanthology.org/2026.findings-acl.1336/.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. *arXiv preprint arXiv:2305.20050*, 2023. URL https://arxiv.org/abs/2305.20050.

Jingwei Ni, Ekaterina Fadeeva, Tianyi Wu, Mubashara Akhtar, Jiaheng Zhang, Elliott Ash, Markus Leippold, Timothy Baldwin, See-Kiong Ng, Artem Shelmanov, and Mrinmaya Sachan. ReProbe: Efficient test-time scaling of multi-step reasoning by probing internal states of large language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pages 11667–11689. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.acl-long.536. URL https://aclanthology.org/2026.acl-long.536/.

Yucheng Sun, Alessandro Stolfo, and Mrinmaya Sachan. Probing for arithmetic errors in language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 8111–8128. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.411. URL https://aclanthology.org/2025.emnlp-main.411/.

An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, Keming Lu, Mingfeng Xue, Runji Lin, Tianyu Liu, Xingzhang Ren, and Zhenru Zhang. Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024. URL `https://arxiv.org/abs/2409.12122`.

Aojie Yuan, Zhiyuan Julian Su, Haiyue Zhang, Yi Nian, and Yue Zhao. Hidden error awareness in chain-of-thought reasoning: The signal is diagnostic, not causal. *arXiv preprint arXiv:2605.09502*, 2026. URL `https://arxiv.org/abs/2605.09502`.

Chujie Zheng, Zhenru Zhang, Beichen Zhang, Runji Lin, Keming Lu, Bowen Yu, Dayiheng Liu, Jingren Zhou, and Junyang Lin. ProcessBench: Identifying process errors in mathematical reasoning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, pages 987–1011. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.50. URL `https://aclanthology.org/2025.acl-long.50/`.

A Data accounting

Table 3: Attempted and retained traces and boundaries by frozen partition.

Partition	Attempted	Retained	Excluded	Boundaries	Error traces	Correct traces
Train	2,041	2,014	27	14,944	1,311	703
Validation	681	677	4	4,980	444	233
Test	678	669	9	4,985	432	237
All	3,400	3,360	40	24,909	2,187	1,173

Table 4: Attempted, retained, and excluded traces by source and frozen partition.

Partition	GSM8K	MATH	OlympiadBench	Omni-MATH
Train	236/236/0	597/592/5	602/594/8	606/592/14
Validation	80/80/0	198/198/0	204/202/2	199/197/2
Test	84/84/0	205/204/1	194/188/6	195/193/2

The retained boundary counts by source are 1,229, 3,718, 5,129, and 4,868 for train; 414, 1,303, 1,708, and 1,555 for validation; and 439, 1,387, 1,661, and 1,498 for test, in the source order shown in the table. The original attempted total was 25,697 boundaries. The retained total is 24,909 after excluding 40 traces that exceed the context limit.

B Descriptive subgroup audit

Table 5: Trace-macro outcomes by ProcessBench source with 95% whole-trace bootstrap intervals.

Source	Complete accuracy	Exact error	Correct rejection
GSM8K	.512 [.405,.619]	.238 [.116,.375]	.786 [.658,.902]
MATH	.500 [.426,.569]	.347 [.264,.435]	.723 [.620,.818]
OlympiadBench	.298 [.234,.362]	.207 [.138,.282]	.463 [.340,.585]
Omni-MATH	.358 [.290,.425]	.324 [.253,.401]	.467 [.319,.614]

Among traces with an incorrect final answer, complete-trace accuracy is 0.292 [95% CI $[0.245, 0.342]$]. Among traces with a correct final answer, it is 0.506 [95% CI $[0.451, 0.554]$]. Across generators with at least 20 held-out traces, complete-trace accuracy ranges from 0.316 to 0.621. These are descriptive subgroup results, not generator rankings or independent hypothesis tests.

C Reproduction record

The maintained package uses one configuration and one Click command group. The principal commands are:

```
uv sync --extra dev
uv run math-error --config configs/project.yaml validate-config
uv run math-error --config configs/project.yaml run-all
uv run math-error --config configs/project.yaml analyze
uv run math-error --config configs/project.yaml fit-conditional
uv run math-error --config configs/project.yaml fit-transition
uv run math-error --config configs/project.yaml transition-diagnostics
uv run pytest
uv run ruff check src tests
```

The project configuration records seed 42, the model and dataset identities, the 2,048-token limit, selection grids, bootstrap counts, and artifact paths. The extraction and analysis artifacts record selected layers, thresholds, predictions, metric summaries, bootstrap intervals, and the data flow. The current repository also includes the unified package layout and neutral test layout described in `README.md`.

Table 6: Compute and software record.

Activation extraction	One NVIDIA A100 GPU, bfloat16, batch size 16, single causal passes over the retained traces.
Probe fitting and audits	CPU-capable; the control audit uses 2,000 whole-trace bootstrap draws.
Software record	Python 3.10 or newer; dependencies are declared in `pyproject.toml` and resolved in `uv.lock`.
Runtime record	The CPU audit records Python, NumPy, pandas, SciPy, scikit-learn, matplotlib, and platform versions in its `summary.json`.
Identifiers	The extraction record fixes Qwen2.5-Math-1.5B-Instruct, bfloat16 dtype, the 2,048-token limit, and the dataset SHA-256.
Licensing	Dataset and model licensing follows the published terms of the corresponding sources.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Introduction name the single model and benchmark, distinguish the persistent prefix target from onset detection, and state the negative and post-hoc results.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The final main-text section discusses one-model scope, shortcut sensitivity, post-hoc analyses, the absence of preregistration, and the invalid causal assay.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains no theoretical results or proof claims.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 and Appendix D specify the split, target, selection protocol, seeds, locked package versions, dataset identity, and exact reproduction commands. The joint shortcut controls share the hidden probe's validation budget, and the transition-matching protocol is stated with its reuse diagnostics.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: An anonymized code supplement includes the pipelines, tests, configurations, and instructions; the upstream dataset and model are publicly accessible.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 3 reports the data partitions, exclusions, model and activation location, probe family, hyperparameter grid, threshold selection, controls, and estimands.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The main results use 95% whole-trace bootstrap intervals, paired control differences, and a two-way clustered bootstrap for reused transition controls.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Appendix D, Table 6 records the GPU model, precision, batch size, approximate per-stage compute, and storage footprint. Exact wall-clock logs for the original extraction were not preserved; the paper states the order of magnitude instead of inventing precision.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work analyzes publicly released model and benchmark artifacts, makes bounded claims, preserves submission anonymity, and reports failures and limitations.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

Justification: The responsible-use discussion identifies research-diagnostic value and the concrete risks of false alarms, late detections, and distribution shift in grading uses.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper releases no new pretrained model, scraped dataset, or other asset with a high misuse risk; it releases analysis code for public research artifacts.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Appendix D, Table 6 records the Apache 2.0 licenses of the ProcessBench dataset and the Qwen model, and notes that the benchmark's constituent sources retain their original terms; the paper also cites both upstream artifacts.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.

- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The anonymous supplement documents the analysis code, resolved configurations, artifact schemas, commands, tests, and known limitations.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: This study performs no crowdsourcing and recruits no human subjects.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: This study performs no human-subject research.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: The analyzed language model is the experimental subject, not a non-standard agent used to generate labels, hypotheses, or core methodological decisions. Agent assistance contributed to code, analysis, and manuscript editing under author direction; the authors verified all reported numbers against repository artifacts and remain responsible for correctness and originality.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.